

Fig. 4.6 Rigid representation of a water molecule used in TIP4P, indicating the "ghost" charge site

sets of whose charge sites are denoted by A and B, takes the form

$$\varphi^{\text{water}-\text{water}} = \sum_{i\in A}\sum_{j\in B} k_1 \frac{Q_i Q_j}{r_{ij}} + \frac{k_2}{r_{\text{OO}}^{12}} - \frac{k_3}{r_{\text{OO}}^{6}},$$

where r_{OO} represents the distance between the oxygen atoms of the two molecules, and k_1, k_2, k_3 are positive constants. Similar potentials are needed for interactions between the atoms of the solute and the atoms of the water molecules.

More generally, consider a collection of N atoms (masses m_i, positions $\boldsymbol{q}_i$, $i = 1, 2, \ldots, \text{N}$) subject to an external potential U. The kinetic energy is

$$K = \frac{1}{2}\sum_{i=1}^{\text{N}} m_i \|\dot{\boldsymbol{q}}_i\|^2.$$

The atomic positions may be referred to a coordinate system placed at the center of mass of the collection

$$\boldsymbol{q}_{\text{cm}} = \frac{\sum_{i=1}^{\text{N}} m_i \boldsymbol{q}_i}{M},$$

where $M = \sum_{i=1}^{\text{N}} m_i$ is the total mass. We write $\boldsymbol{q}_i = \boldsymbol{q}_{\text{cm}} + \boldsymbol{\delta}_i$, where $\sum_{i=1}^{\text{N}} m_i \boldsymbol{\delta}_i = 0$. Thus the kinetic energy becomes

$$\begin{aligned} K &= \frac{1}{2}\sum_{i=1}^{\text{N}} m_i \|\dot{\boldsymbol{q}}_{\text{cm}}\|^2 + \sum_{i=1}^{\text{N}} m_i \dot{\boldsymbol{q}}_{\text{cm}} \cdot \dot{\boldsymbol{\delta}}_i + \frac{1}{2}\sum_{i=1}^{\text{N}} m_i \|\dot{\boldsymbol{\delta}}_i\|^2 \\ &= \underbrace{\frac{M}{2}\|\dot{\boldsymbol{q}}_{\text{cm}}\|^2}_{\equiv K_{\text{trans}}} + \underbrace{\frac{1}{2}\sum_{i=1}^{\text{N}} m_i \|\dot{\boldsymbol{\delta}}_i\|^2}_{\equiv K_{\text{rot}}}, \end{aligned}$$

where we have used the fact that $\sum_{i=1}^{\text{N}} m_i \boldsymbol{\delta}_i \equiv 0$, hence that the time derivative also vanishes, in order to remove the middle term. At time t, each vector $\boldsymbol{\delta}_i$ can be related to its initial value by a certain orthogonal 3×3 rotation matrix $\boldsymbol{\Theta}(t)$: